CS321 Machine Learning, Data Mining and Business Analytics

TITLE: INCOME FORECASTING USING GUIDED LEARNING BASED ON THE UCI ADULT DATABASE.

COURSE: MACHINE LEARNING, DATA MINING & BUSINESS ANALYTICS.

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1. Introduction

The goal of this project is to predict whether a person earns more than $50,000 per year based on demographic characteristics and employment based on UCI adult income data. We use three guided learning algorithms - Decision Tree, Random Forest, and Reverse hardware - to compare performance, clarity, and practical applications.

Introduction

Dataset description

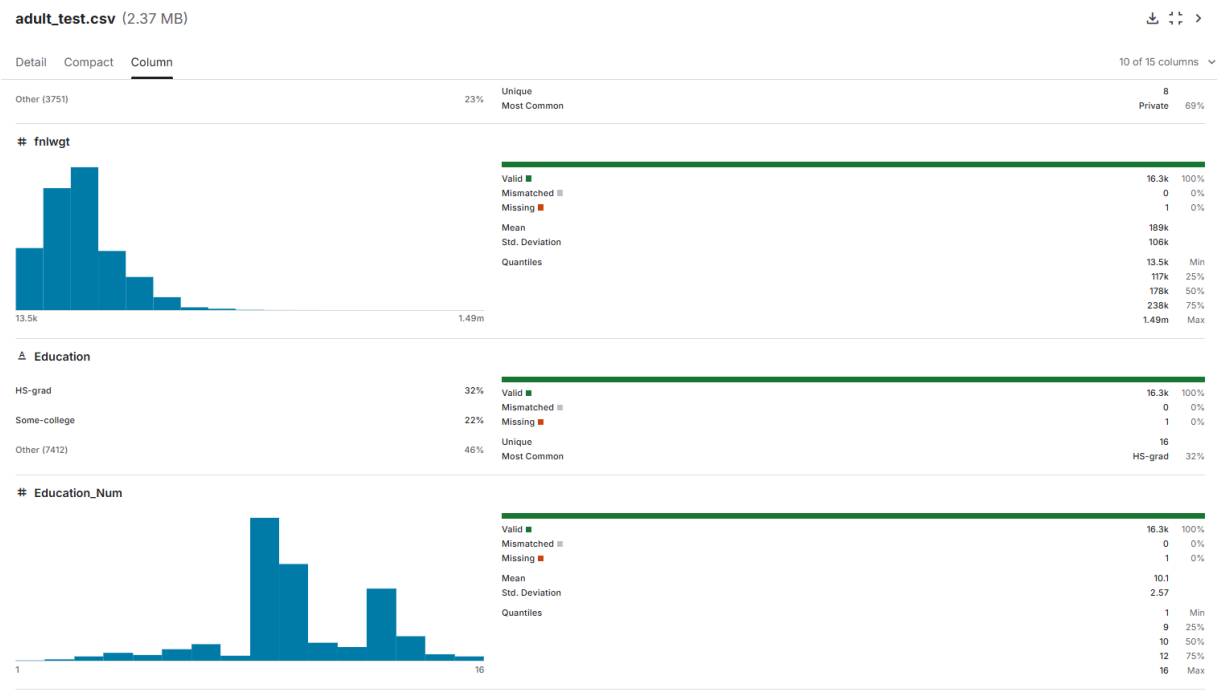
Pre-processing of data

Models development

Model Evaluation

Interpretability and practical implications

2. Dataset description



- Source: UCI Machine Learning Repository - Adult Income Dataset
- Data Number of events: ~ 48,842 records
- Specifications: 14 attribites:
- Quantity: age, fnlwgt, education-num, hours-per-week, capital-gain, capital-loss
- Category: workclass, education, marital-status, occupation, relationship, race, gender, native-country, income
- Target: income (binary: “<=50K” or “>50K”)
- Distribution: imbalanced (~76% ≤50K, ~24% >50K)

3. Pre-processing of data

- Replaced the missing values (marked as "?") with NaN.
- We filled in the missing categorical values using the imputation mode for the working class, profession, and native country.
- Removed spaces from string entries.
- Encoded all categorical variables using label encoding (via sklearn.preprocessing.LabelEncoder).
- Retained numerical characteristics (age, number of hours per week) as it is.
- Checked that there were no missing values after preprocessing.
- Divide the shuffled data into:
- Training set: 70%
- Test set: 15%
- Test set: 15%

4.Models development

We have implemented three classifiers using scikit-learn:

All models were trained on a training set and the accuracy was measured during the development process.

Model	Key Parameters
Decision Tree	criterion='gini', max_depth=10, random_state=42
Random Forest	n_estimators=100, max_depth=15, n_jobs=-1, random_state=42
Logistic Regression	C=1000 (low regularization), penalty='l2', max_iter=10000, random_state=15

5. Model Evaluation

We evaluated each model in the test set using standard classification indicators:
Performance Summary (Test Set)

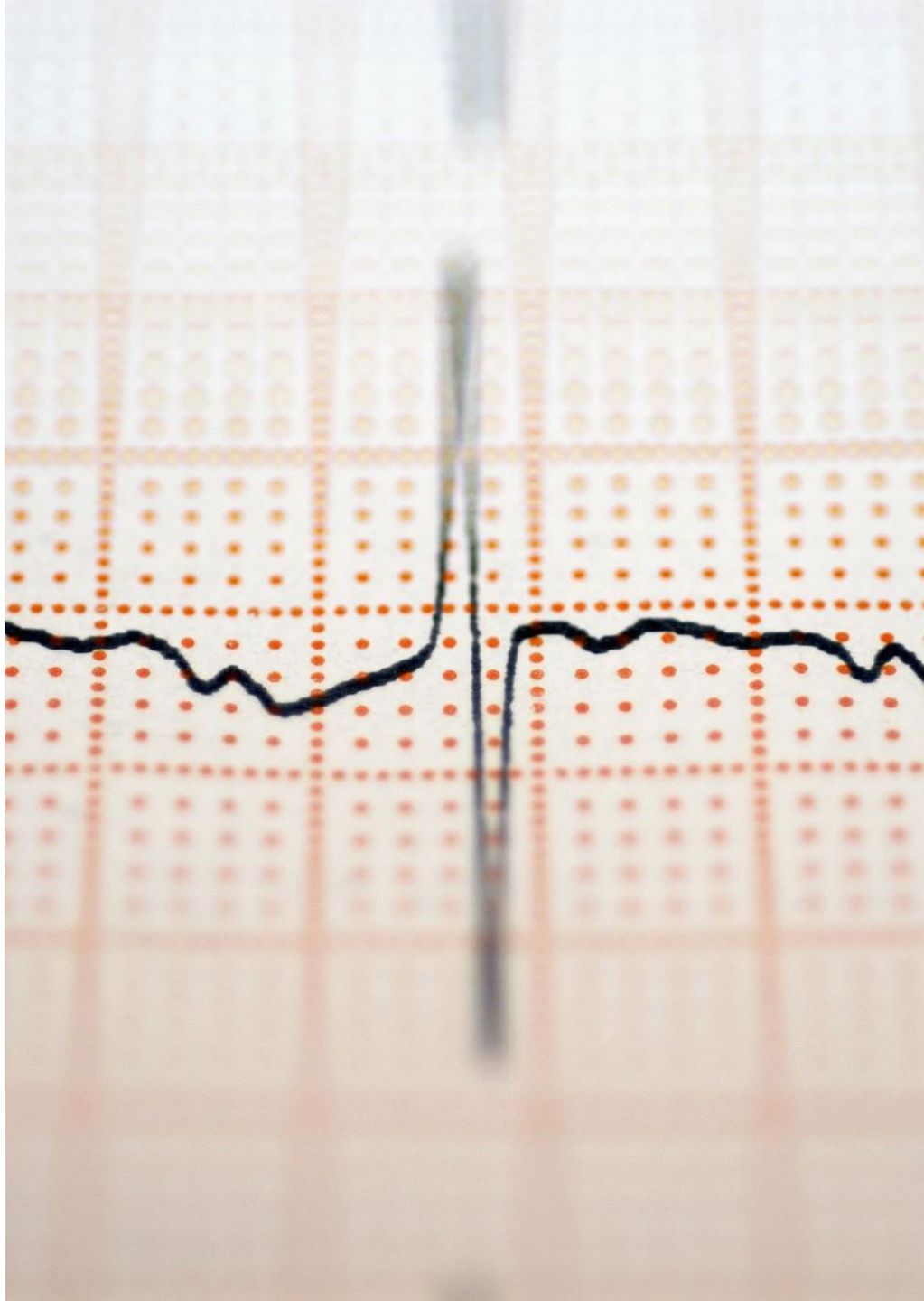
	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
0	Decision Tree	0.824894	0.648734	0.584712	0.615062	0.851017
1	Random Forest	0.836086	0.685734	0.581289	0.629207	0.884578
2	Logistic Regression	0.764160	0.521441	0.173417	0.260274	0.761604

Key Observations

- Random Forest has achieved the highest performance across all metrics thanks to ensemble averaging and reduced overfitting.
- The decision tree showed good interpretability, but a slightly lower degree of generalization than Random Forest.
- Logistic regression performed poorly due to non-linear relationships in the dataset and categorical complexity, although it could still be interpreted using coefficients.

Visualizations

- The confusion matrices confirmed that all models had higher false negative results (prediction <50,000 with true income >50,000), reflecting a class imbalance.
  See Appendix 1.
- ROC curves showed strong discriminative ability (AUC > 0.87 for all models).
  See Appendix 2.
- Feature importance from the Decision Tree revealed **age**, **hours-per-week**, and **occupation** as top predictors—aligning with socioeconomic intuition.
  See Appendix 3.

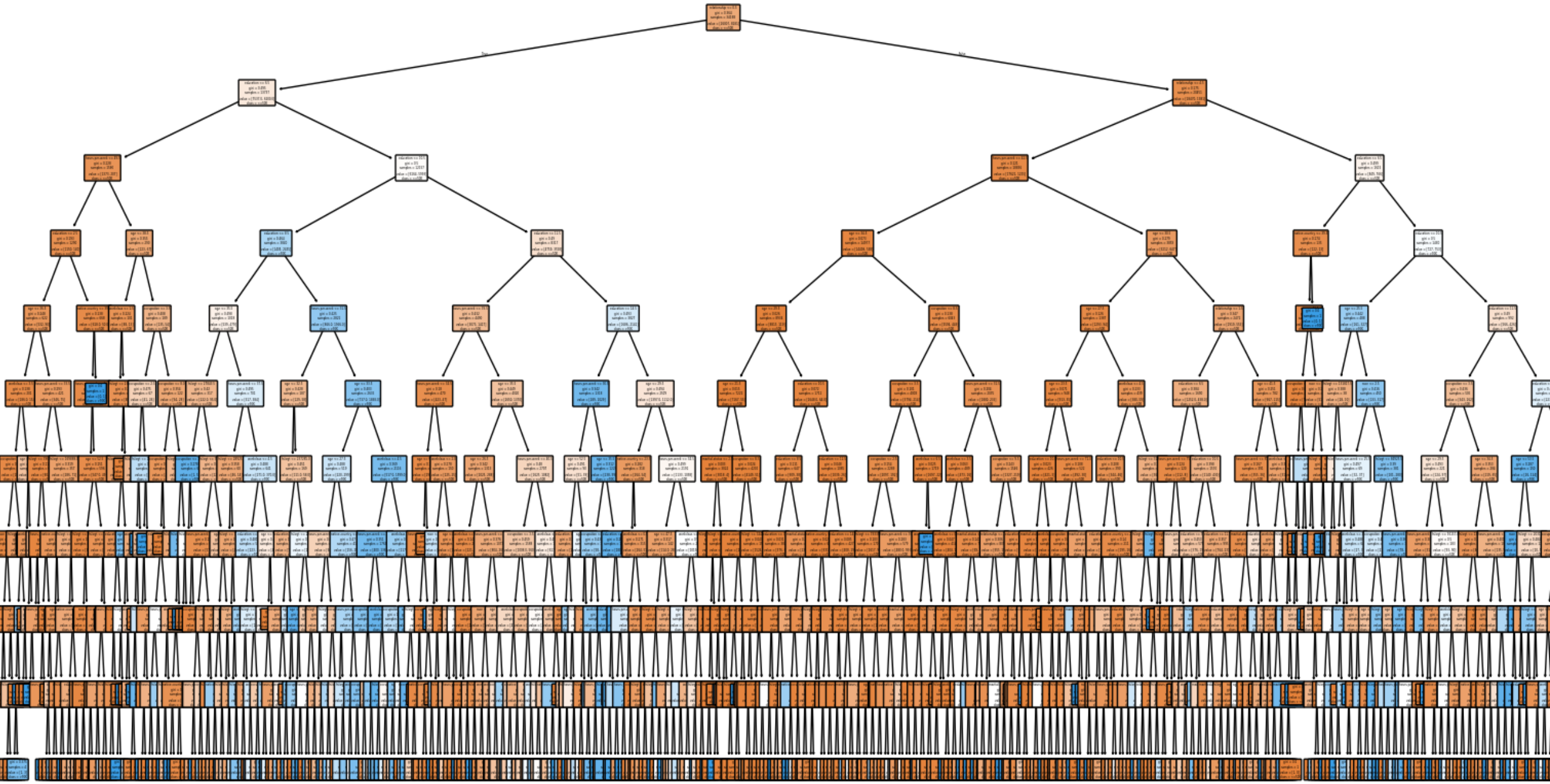


6. Interpretability and practical implications

The Importance function shows which factors have the greatest impact on revenue forecasting, which is useful for bias auditing (for example, to check whether race or gender have an undue influence).

Understanding Business: Employers or policy makers can use such models to identify risk groups or assess wage equality, if equity constraints are met.

Decision Tree Classifier – Income Prediction



Cross-Validation

5-fold CV confirmed stable performance

Decision Tree CV accuracy:
0.8253 ± 0.0017

Random Forest CV accuracy:
0.8374 ± 0.0032

Logistic Regression CV Accuracy:
0.7625 ± 0.0015

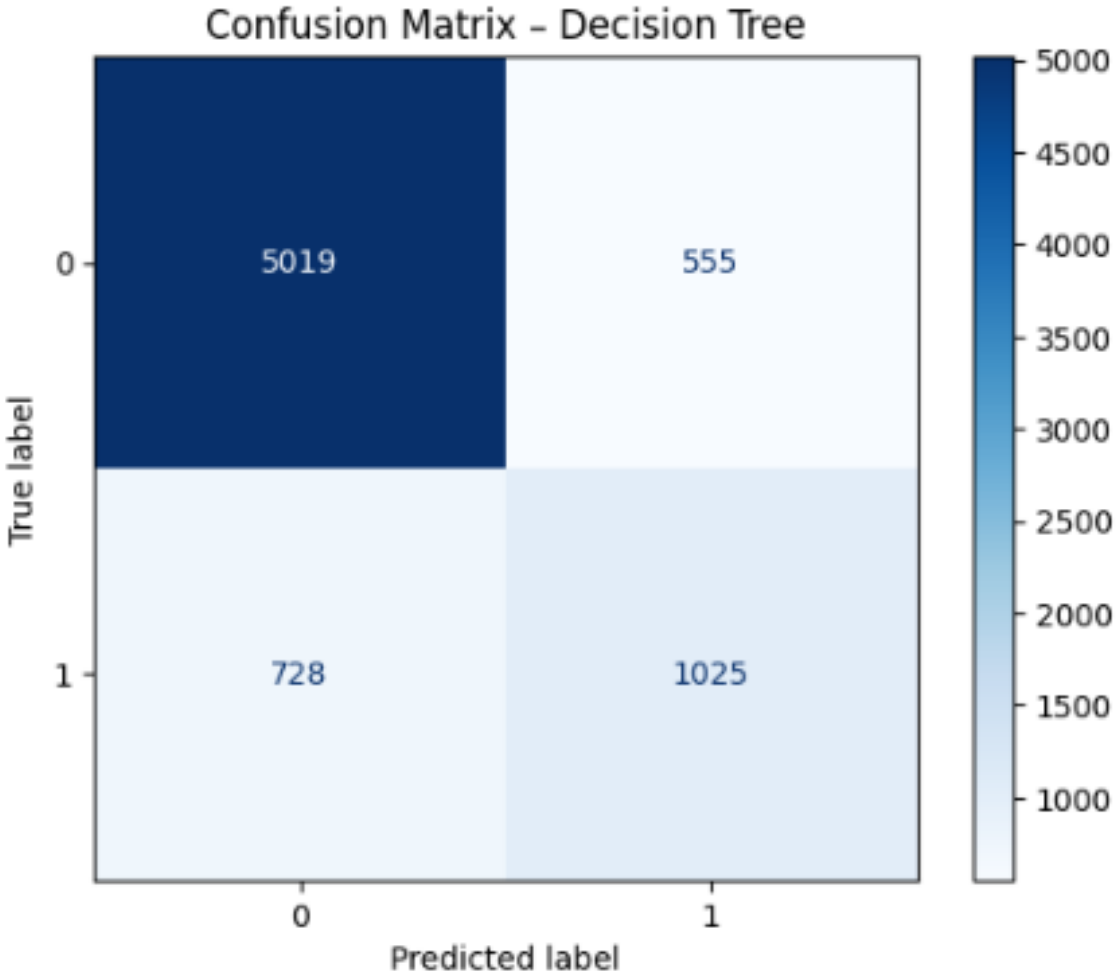
7. Conclusion

- Our analysis shows that group methods such as random forests offer the best compromise between accuracy and reliability to solve this problem. A decision tree provides valuable interpretability, while logistic regression provides a reliable linear framework. All models confirm that income is strongly correlated with the length of the working day, occupation, and age, which is confirmed by the real labor economy.

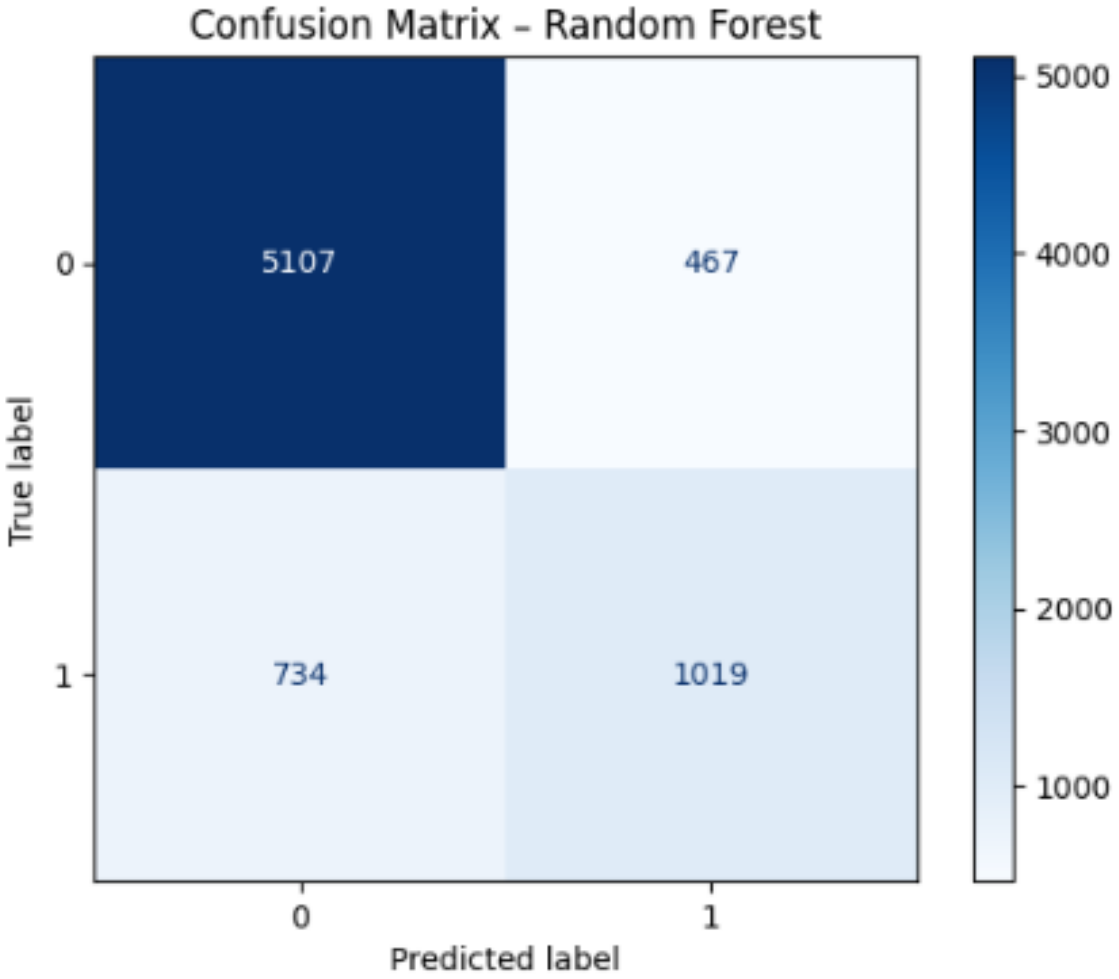
8. Appendix

- **Code**: implementation in Google Colab https://colab.research.google.com/drive/18M9CAsTtsJXLH4fLdIY6bxnIrF3sbPuQ#scrollTo=lsqnUHAoD1CQ
- **Figures**:
  - Confusion matrices for all models (Appendix 1)
  - ROC curve comparison (Appendix 2)
  - Feature importance bar chart (Appendix 3)
  - Decision Tree visualization (Appendix 4)
- **Dataset**: UCI Adult Income Dataset (https://storage.yandexcloud.net/uol/programming/adult.csv

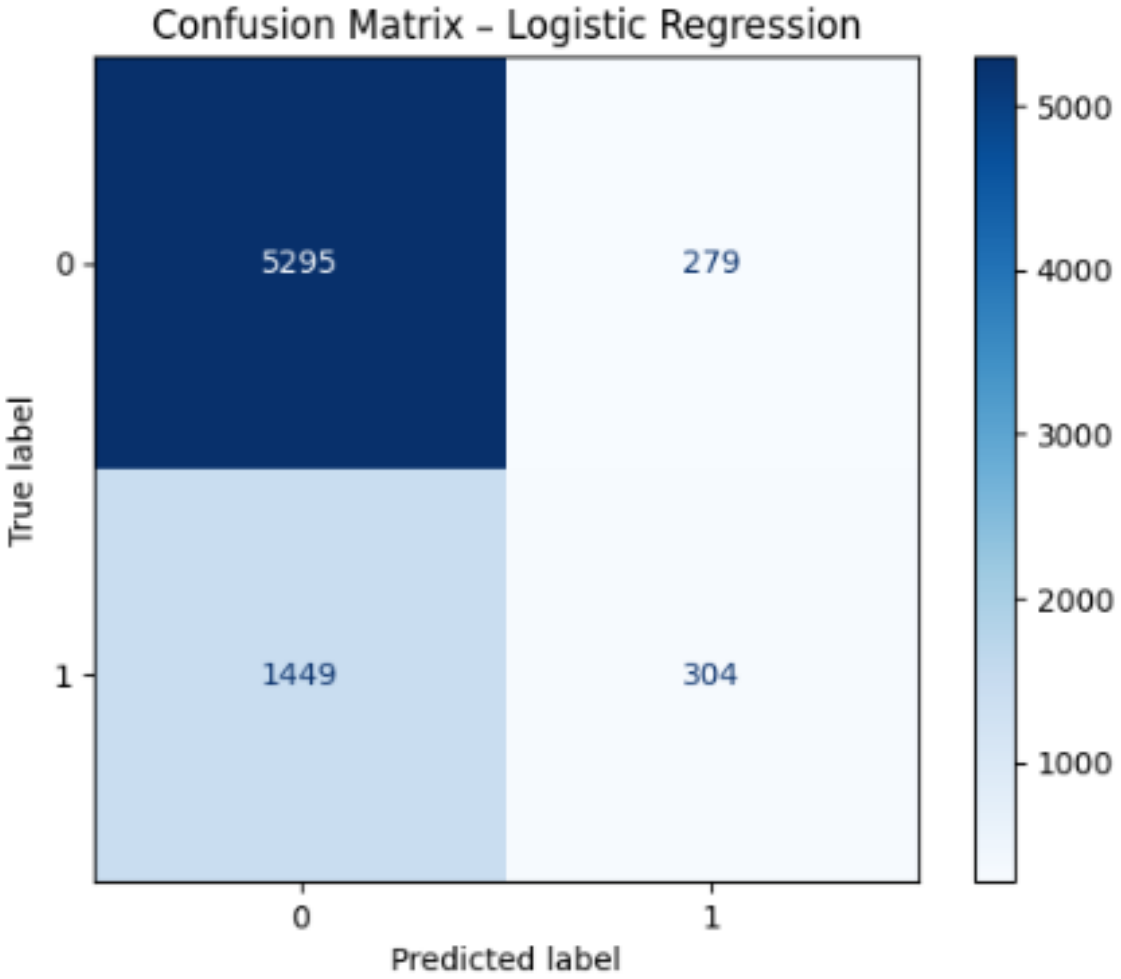
Appendix 1.1


Confusion Matrix – Decision Tree
5019
555
728
1025
0
1
True label
0
1
Predicted label
5000
4500
4000
3500
3000
2500
2000
1500
1000

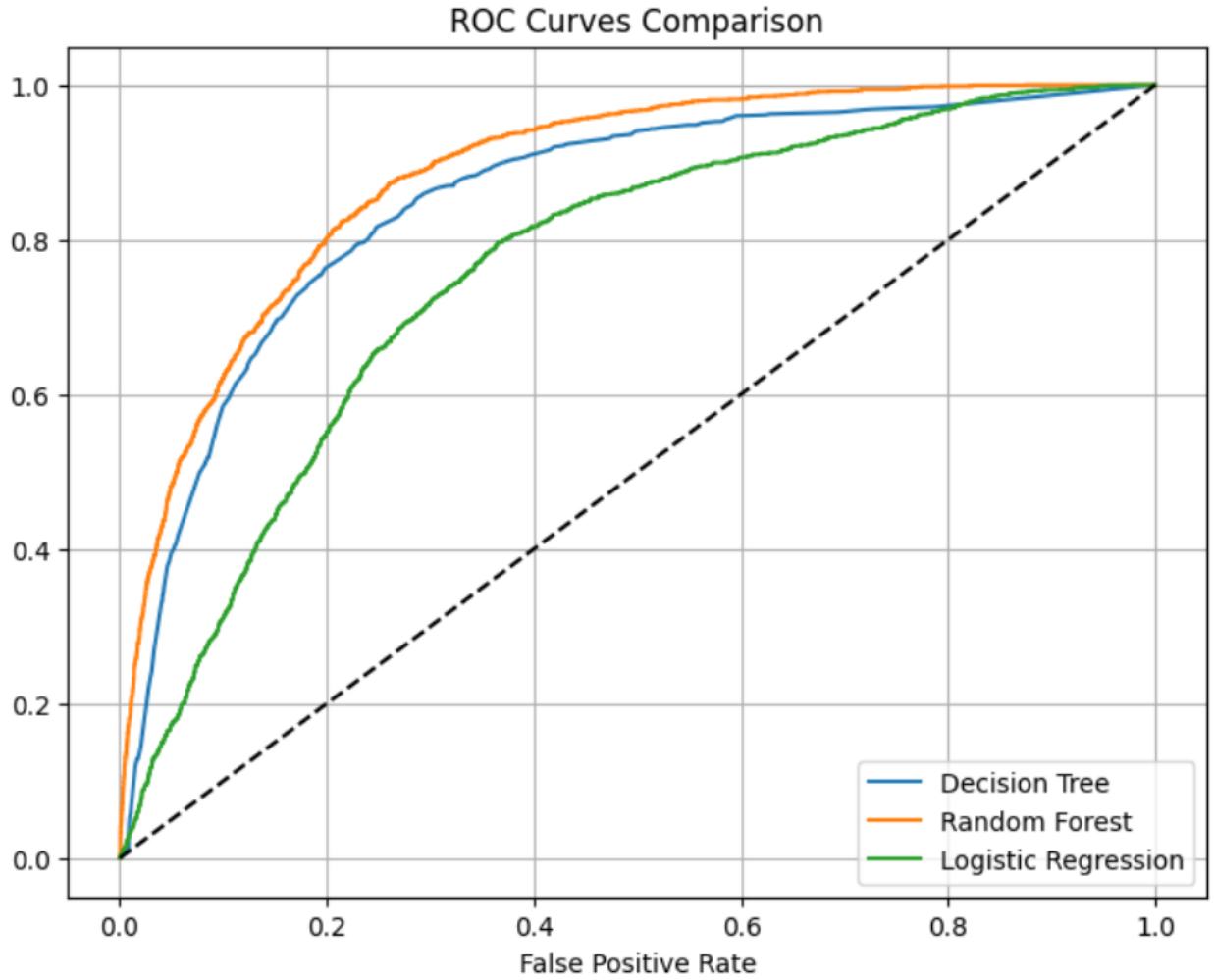
Appendix 1.2

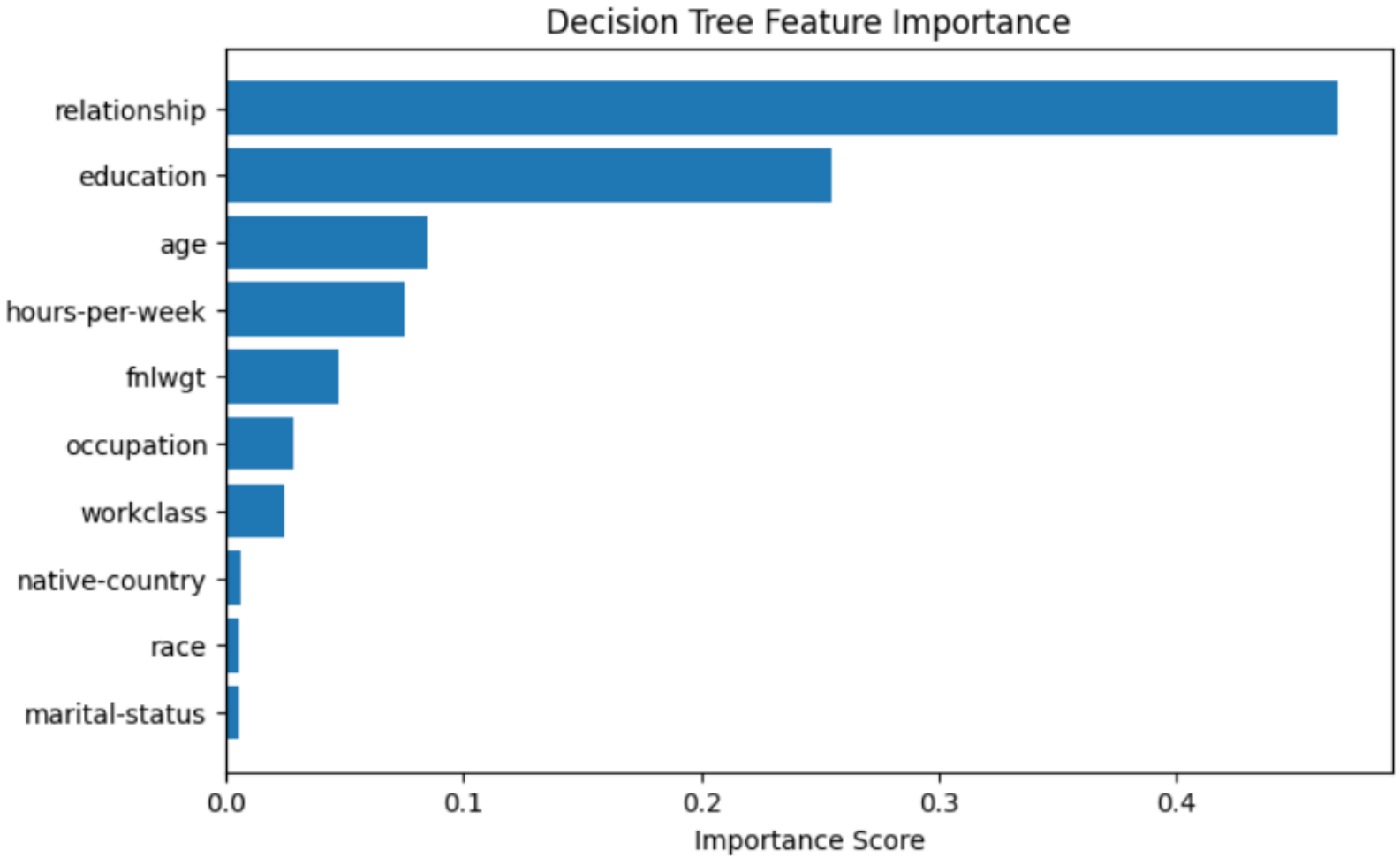

Confusion Matrix – Random Forest
5107
467
734
1019
0
1
True label
0
1
Predicted label
5000
4000
3000
2000
1000

Appendix 1.3


Confusion Matrix – Logistic Regression
5295
279
1449
304
0
1
True label
0
1
Predicted label
5000
4000
3000
2000
1000

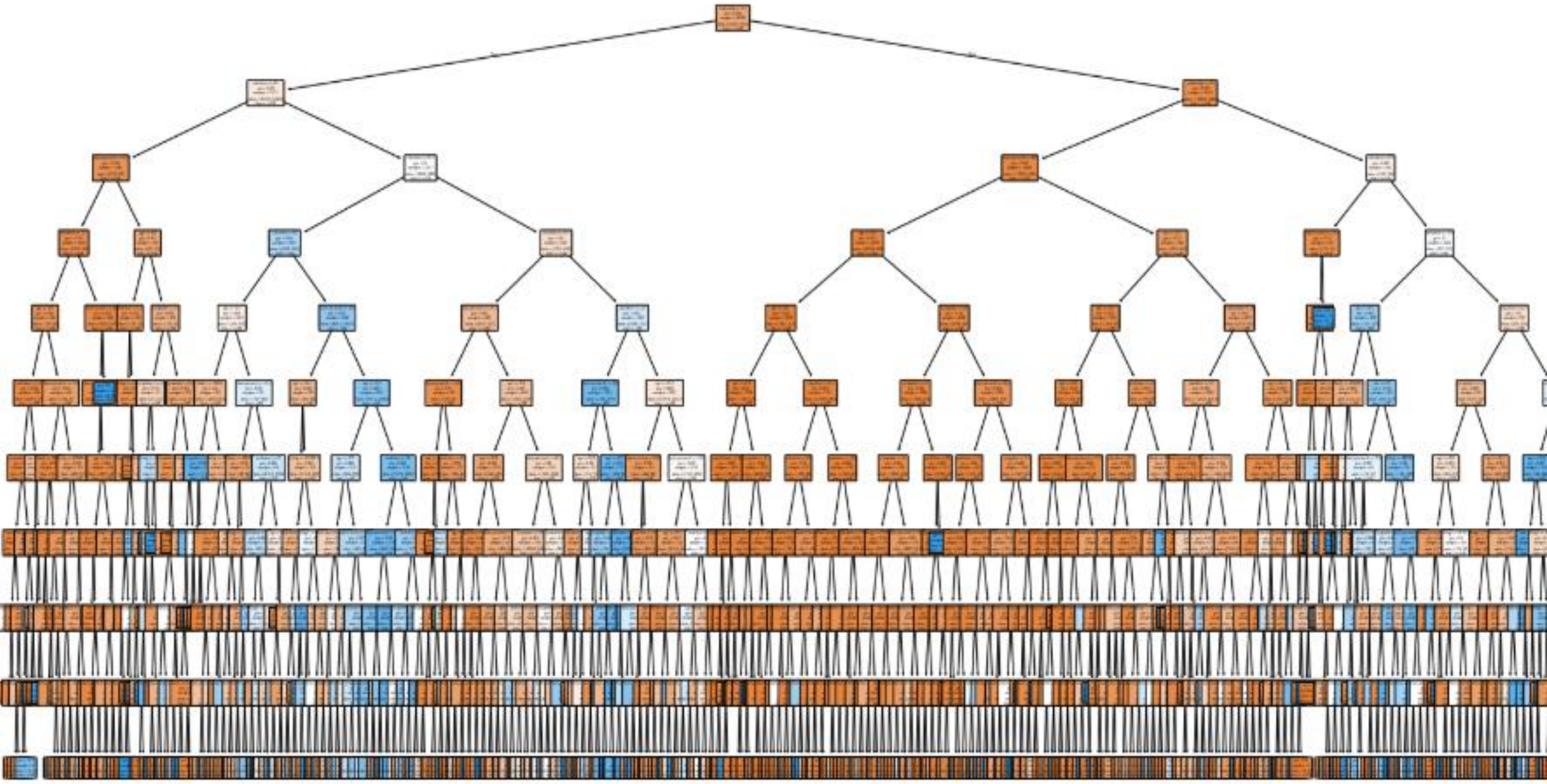
Appendix 2


ROC Curves Comparison
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
False Positive Rate
Decision Tree
Random Forest
Logistic Regression


Decision Tree Feature Importance
relationship
education
age
hours-per-week
fnlwgt
occupation
workclass
native-country
race
marital-status
0.0
0.1
0.2
0.3
0.4
Importance Score

Appendix 3

Appendix 4

- Also, you can see it here: https://storage.yandexcloud.net/uol/image1.png

Decision Tree Classifier – Income Prediction